

# STATEMENT OF NEED (SON)

## 1.0 TITLE

Acquisition of a Synergy H1 Hybrid Multi-Mode Reader

## 2.0 BACKGROUND

The Pediatric Oncology Branch (POB), NCI, NIH is dedicated to improving outcomes for children and young adults with cancer and genetic tumor predisposition syndromes through translational research spanning basic science to clinical trials. The Branch consists of 10 research programs, all of which rely on shared laboratory instrumentation to support their scientific and clinical research activities.

Currently, the POB has a single plate reader (SpectraMax M3) that supports all 10 programs. This instrument is approaching 12 years of age, has limited manufacturer support, and represents a single point of failure with no available backup. Continued reliance on this aging instrument poses a significant risk to ongoing research operations.

The purpose of this order is to purchase a new multimode plate reader to ensure continuity of research, support all 10 programs, and provide a reliable, modern platform capable of meeting current and future assay requirements.

## 3.0 TYPE OF ORDER

This is a Firm Fixed-Price Purchase Order.

## 4.0 SPECIAL ORDER REQUIREMENTS

### 4.1 PRODUCT FEATURES/SALIENT CHARACTERISTICS

The following salient characteristics are required to ensure the instrument can support diverse assay types, enable future assay development, and provide reliable long-term operation across multiple research programs. The following product features/characteristics are required for this requirement:

Agilent BioTek Synergy H1 Hybrid Multi-Mode Reader, Model: H1M Part# SH1M-SN (Qty=1)

- **Core Detection Capabilities**
  - Must support **absorbance, fluorescence, and luminescence** detection in a single instrument
  - Must provide **monochromator-based wavelength selection** for absorbance and fluorescence, enabling flexible assay design
  - Must support **full spectral scanning** for both excitation and emission wavelengths
- **Performance and Sensitivity**
  - Must provide **high sensitivity for fluorescence and luminescence assays** suitable for low-signal applications
  - Must support **broad dynamic range** to accurately quantify signals across diverse assay types

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- **Fluorescence Capabilities**
  - Must support **top and bottom fluorescence detection** to accommodate both solution-based and cell-based assays
  - Must be compatible with **modern fluorophores** used in advanced biological assays
- **Assay Flexibility**
  - Must support a wide range of assays, including:
    - ELISA and colorimetric assays
    - Fluorescent reporter assays
    - Luminescent viability and gene expression assays
    - Nucleic acid and protein quantification
  - Must enable **assay development and optimization**, including the ability to adjust excitation/emission wavelengths
- **Modularity and Future Expansion**
  - Must provide a **modular design** allowing for future upgrades (e.g., additional detection modes such as TRF or FP)
  - Must support evolving research needs without requiring replacement of the base instrument
- **Plate and Sample Compatibility**
  - Must accommodate **standard microplate formats (6- to 384-well plates)**
  - Must support **microvolume sample analysis** (e.g., DNA/RNA quantification)
- **Environmental and Operational Features**
  - Must include **temperature control and incubation capabilities** for kinetic and cell-based assays
  - Must support **orbital shaking** for assay mixing and uniformity
  - Must be compatible with **automated workflows and standard laboratory software systems**
- **Service and Support**
  - Must have **ongoing manufacturer support and service availability**
  - Must provide access to **software updates and technical support**
- **Physical and Space Requirements**
  - The instrument must have a **compact benchtop footprint compatible with available laboratory space**, allowing installation without requiring renovation or reconfiguration of existing lab infrastructure

## 4.2 DELIVERY / INSTALLATION

Item shall be delivered within 60 days of purchase order award, etc.)